PDF Accessibility Assessment Using Multimodal Deep Learning

Project Proposal

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1. Problem and Motivation

For many instructors, complying with ADA guidelines is a difficult task and can limit creative freedom when creating content for online courses. One major problem area is PDF documents, which can be difficult to modify and remediate without repeatedly checking them against Web Content Accessibility Guidelines (WCAG) 2.0. Instructors may not have enough time, or may find it too difficult, to create PDF content while also ensuring that it meets accessibility guidelines. As a result, documents may be removed from courses and instructors may have to redo work. This affects students as well as instructors: students with and without disabilities need course content to remain readily available. Removing inaccessible PDFs can create a significant disruption to learning.

2. Proposed Technology and Approach

To reduce instructors' workload while helping ensure that accessible content is available to students, this project proposes a website that accepts uploaded PDFs, analyzes their contents, identifies potential accessibility problems, and produces an interpretable accessibility score.

The main model architecture will be a multimodal Transformer. Microsoft's LayoutLMv3 is a candidate architecture because it combines textual, visual, and spatial information. This allows the model to learn relationships among text, images, and their positions on a PDF page. For example, text containing the word "Introduction," appearing near the top of a page in a larger font, may be recognized as a heading based on its text and layout context.

LayoutLMv3 is pretrained and will be adapted to the project through transfer learning. The model will first learn to recognize PDF elements and then be fine-tuned for selected accessibility-related classifications. Potential predictions include missing alternative text, incorrect heading structure, and no detected issue. Predictions will be associated with document elements such as headings, images, and tables. The system may also report confidence values for its predictions.

The model's predictions will be combined with programmatic PDF inspection. This distinction is important because some accessibility properties—such as whether an image has alternative text or whether document language metadata is present—cannot reliably be inferred from the rendered page alone. Python libraries will be used to extract and inspect PDF content and structure, including PyMuPDF and pikepdf, alongside Hugging Face Transformers and LayoutLMv3.

For each selected WCAG criterion, the system will produce a result such as Pass, Fail, or Needs Review. An overall accessibility percentage will then be calculated from the applicable criterion-level results. This percentage will be a project-defined assessment metric, not an official WCAG conformance rating. Criteria that do not apply to a document will be excluded from the denominator, and uncertain results will remain separate from passes and failures.

3. Datasets and Data Augmentation

The project will use two primary datasets: DocLayNet and the PDF Accessibility Benchmark (2025). DocLayNet contains more than 80,000 manually annotated document pages and labels layout elements such as titles, section headers, text, images, tables, and lists. It is available through GitHub and Hugging Face. The PDF Accessibility Benchmark (2025) contains 125 PDF documents with criterion-level annotations, including Passed, Failed, Not Present, and Cannot Tell labels. It is available through GitHub.

Because the accessibility benchmark is relatively small, data augmentation may be necessary. One strategy is to create controlled variants of existing PDFs by introducing specific accessibility issues—for example, removing alternative text or changing heading or reading-order tags. A single source PDF could produce multiple variants, each with a known modification and corresponding label. The modifications will be limited so that the document remains meaningful and the intended accessibility issue is isolated.

All variants derived from the same source PDF will be kept within the same training, validation, or test split. This will help prevent data leakage and overly optimistic evaluation results. Visual transformations, such as mild rotation or changes in scan quality, may also be used to improve robustness, provided that they do not change the ground-truth accessibility labels.

4. Training and Evaluation Plan

First, the pretrained LayoutLMv3 model will be fine-tuned on DocLayNet to recognize and classify document elements. The resulting document-understanding model will then be adapted to accessibility-related tasks using the PDF Accessibility Benchmark and carefully labeled augmented examples. The system will combine model predictions with PDF-level structural checks to produce criterion-level Pass, Fail, or Needs Review outcomes.

The accessibility score will be calculated from the criterion-level results rather than predicted directly by the neural network. Model performance will be evaluated using appropriate classification measures, such as precision, recall, and F1-score, and by comparing predictions with the benchmark annotations. Evaluation will use document-level data splits so that pages or augmented versions of the same original PDF do not appear across training and testing.

5. Innovation

The main contribution of this project is its combination of document understanding and WCAG-focused PDF analysis. Rather than simply detecting document elements or producing an unexplained score, the proposed system will use multimodal document representations to identify elements and potential accessibility issues, then combine those predictions with structural PDF verification.

The use of transfer learning is another important aspect. Training on DocLayNet will provide the model with prior knowledge of common document elements and layouts before it is adapted to the smaller accessibility dataset. The project will investigate whether this prior document-layout knowledge improves the identification of accessibility issues when labeled accessibility data is limited.

The project will also explore controlled synthetic data generation as a way to address the shortage

of accessibility-labeled PDFs. Finally, criterion-level results will make the overall score more interpretable by showing which accessibility checks passed, failed, or require human review.

6. Anticipated Challenges and Risk Mitigation

The primary challenge is the limited size of the PDF Accessibility Benchmark, which is the most directly relevant dataset for the project. Although controlled augmentation can create additional examples, developing and validating these examples may be time-consuming. To mitigate this risk, the project will focus on a small number of selected WCAG criteria, prioritize a manageable set of controlled PDF modifications, use transfer learning, and monitor class balance. Augmented documents will be grouped with their originals when creating data splits.

A second challenge is that accessibility cannot always be determined from a PDF's visual appearance. For example, an image may look correct but lack alternative text, or a visually prominent heading may not be tagged semantically. The system will therefore combine LayoutLMv3 with PDF structural inspection. Issues that cannot be reliably determined automatically will be marked Needs Review rather than assigned an unsupported result.

A third challenge is generalization. The initial project will focus on text-based PDFs, while real course materials may include scanned documents, slide-like layouts, and other formats. The model may not generalize equally well across all of them. The project will define a limited target scope, use varied layouts where feasible, and clearly document the types of PDFs and WCAG criteria covered by the prototype.

7. Potential Impact and Feasibility

The project could benefit instructors who need to check PDFs used in their courses, instructional designers who prepare digital learning materials, and accessibility specialists who review course content. Automated preliminary screening could reduce the time spent manually locating potential issues and help instructors prioritize documents that need remediation. Students could benefit from more accessible and consistently available course materials.

The project is moderately feasible within a semester. Using a pretrained model and existing document-layout data reduces the need to train a large model from scratch. The largest time constraints are likely to be preparing accessibility-specific training examples, validating augmented PDFs, integrating structural checks, and evaluating the model. Feasibility will be improved by limiting the initial prototype to a small set of WCAG criteria and treating an interactive website as a stretch goal if time permits.